

Design Opportunity Statement

IIIT Hyderabad Breakfast Mess System

Overview

This is the second of the two deliverables that come out of Activity 8. The first, the Reframed Systemic Problem Statement, diagnoses what the problem actually is. This one converts that diagnosis into a design opportunity, where in the system a change would actually do something, what any intervention has to respect, and what should explicitly not be attempted.

It stops short of designing the interventions themselves. That's Activity 9's job. What this document does is hand Activity 9 a shortlist that is already filtered against the evidence, along with the reasons certain obvious-looking directions have been ruled out, so that Phase 3 doesn't spend effort rediscovering them.

1. Where the Leverage Is, and Isn't

The reframe showed the problem living at the structure and mental-model layers of the Iceberg. That immediately tells us something about where intervening is worth the effort.

Not at the event layer. Persuading individual students to attend, reminders, nudges, campaigns, targets the most visible layer and the least movable one. A meaningful share of non-attendance is legitimate, and the rest is rational given the incentives. Changing behaviour without changing what produces it means fighting the structure permanently.

Not at the parameter layer either, which is where the institution does act when it acts at all. The cancellation cap has already moved from five to ten and back. That kind of change alters where the pressure escapes, into formal cancellation, informal resale, or silent no-shows, without altering whether the pressure exists. Parameters are the easiest thing to change and the least consequential.

At the structural layer: what triggers a bill, and what triggers institutional attention. These are the two rules that produce almost everything else in the diagnosis.

And underneath that, at the mental-model layer: the belief that an aggregate registration count adequately stands in for demand, and that procurement certainty is permanently worth more than demand accuracy rather than a trade-off worth re-making. Changes here are the hardest and the most durable. The encouraging detail is that this institution has already demonstrably suspended both beliefs, twice, under shock conditions, so neither is as fixed as it looks.

2. The Design Opportunity Statement

The opportunity is to give this system a way to notice its own cost.

The Breakfast Paradox persists not because the fix is unknown, it has already been run twice, but because nothing in the system converts the cost of the gap into a signal that reaches anyone with the authority to act on it. Every consequence is absorbed competently before it becomes visible: the resale market takes the financial loss, composting and staff consumption take the surplus, outside venues take the dissatisfaction. The institution's attention triggers on complaint intensity, and this problem is quiet.

The design opportunity is therefore not to build a new solution, but to build the missing feedback: a route by which a known, measured, aggregate cost can travel upward and become a decision. Three intervention points follow directly from the three root causes, at increasing depth, the billing trigger, the ownership gap, and the attention mechanism. The last of these is the cheapest and the most under-considered, and it is the one that would have surfaced the CFS Chair's own figures years ago.

3. The Three Intervention Points

These are stated as opportunities, not designs. Activity 9 works out what each would concretely look like.

3.1 On the rule: what triggers a bill

The institution has already operated attendance-tracked billing successfully during the gas shortage and during holiday periods. The opportunity is a standing, partial version of the model it has already proven, rather than a novel mechanism.

Breakfast specifically is the natural scope. It has the worst turnout of any meal at thirty five to forty five percent, which means it has the most to gain and, by the same token, the least procurement volume at risk in a trial. A change here would be small in absolute terms and large in proportion.

This opportunity is gated on one unanswered question, what the vendor's contract can actually absorb, covered in section 6.

3.2 On decision rights: who owns the outcome

Decision rights over the registration-attendance outcome are split four ways across menu, kitchen execution, academic timetable and billing policy, with no actor holding both the knowledge and the authority. That is the confirmed reason a proven fix was never picked back up.

The opportunity is a single named owner accountable for that outcome end to end. Not a new committee, the system already has escalation structures that demonstrably work when something reaches them. What's missing is an actor for whom this specific gap is their problem, since the diagnosis established that currently it is nobody's.

3.3 On information flow: what gets noticed

The escalation design routes high-intensity feedback to a committee and low-intensity feedback to the CFS Head. Both branches respond to how loudly a problem is felt. Neither responds to how much it costs.

The opportunity is an escalation trigger that fires on measured cost alongside the existing one that fires on complaint volume. This is the cheapest of the three interventions by a wide margin and the one with the broadest reach, because it would also catch the next quiet-but-large problem, not just this one.

It has a precondition: waste is currently never measured by volume, only handled by category. A system with no metric cannot escalate on one. Establishing that measurement is a small change that unlocks the rest.

4. What Any Intervention Must Respect

These are constraints the evidence imposes, not preferences.

Do not remove an absorption mechanism to manufacture a signal. Making resale harder, or ending staff access to leftovers, would make the problem more visible by making real people worse off. Every absorption layer in this system is locally beneficial, students genuinely recover money, waste is genuinely handled responsibly, staff are genuinely fed. The signal the institution needs can be measured directly instead. Any proposal that works by increasing someone's pain to create institutional visibility is the wrong kind of fix, and the diagnosis is what makes that identifiable in advance.

Do not treat turnout as the success metric. A meaningful share of the gap is students eating the way they actually live, intermittent fasting, regional patterns, religious observance. Driving turnout toward a hundred percent would mean engineering away legitimate choices. The right measure is whether the cost becomes visible and answerable, not whether the seats fill.

Do not treat the 8:30 class start as the lever. The academic timetable almost certainly solves real problems of its own, faculty scheduling, room contention, sequencing. Treating it as an obstacle to be removed assumes the academic system's job is to make breakfast easier. It should be treated as a fixed boundary condition the mess system works around, not a variable to pull.

Do not aim the fix at students. They hold the least power and the most defensible behaviour in the entire system. Any intervention whose mechanism is "students should register more accurately" is asking the least-empowered actor to solve a structure they did not build.

Preserve what already works. The Student Council to Committee escalation pathway produced a real policy change on menu rotation. Skip Meal's mechanism is confirmed to reach kitchen forecasting. Neither needs replacing, the first needs a second trigger type, the second needs to be worth a student's time.

5. How the Three Interact

They are not alternatives, and the order matters.

The information-flow change (3.3) is the enabling one. Without a measured cost, there is nothing for an owner to act on and no evidence base for a billing change. It is also the only one that can be done unilaterally by an office that already holds the data.

The ownership change (3.2) is what converts a visible number into a decision. A signal with no addressee produces nothing, which is precisely the current state, the CFS Chair's figures exist and go nowhere.

The billing change (3.1) is the deepest and the one that actually closes the gap, but it is also the one gated on an external constraint. It is best understood as what the first two make possible, not as a standalone move.

A reasonable reading of this is that the sequence runs bottom-up in cost and top-down in impact: measure first, assign an owner second, change the rule third, with the third scoped to breakfast and evaluated rather than assumed.

6. The Question That Gates All of This

Whether the vendor's contract terms could absorb any standing or partial version of the attendance-tracked model has never been asked directly.

This single question determines the shape of the whole opportunity. If the contract can accommodate it, section 3.1 is an adoption problem, the institution would be formalising something it has already run. If it cannot, section 3.1 becomes a renegotiation problem, and the opportunity narrows to changing the specific contractual term that blocks it, which is a different and harder piece of work.

Either way the other two intervention points stand unchanged, which is part of why they are worth pursuing first.

7. What Phase 3 Inherits

Activity 9 asks for leverage points and design principles. This document hands it three leverage points already ranked by depth rather than by ease, five constraints that rule out whole categories of intervention before they are designed, a stated sequencing logic, and one open question that determines whether the deepest of the three is an adoption or a renegotiation problem.

The single most useful thing carried forward is probably the negative result: the reason this system has not fixed itself is not that anyone lacks capability, willingness or information. It is that nothing converts a real, known, aggregate cost into something the system can hear. Designing that conversion is the work.